Project Overview

This project presents a fully automated smart greenhouse monitoring and control system built around the ESP8266 NodeMCU microcontroller. The system continuously reads environmental data from a DHT11 temperature and humidity sensor and a soil moisture sensor, and uses two relay modules to drive an exhaust fan and a water pump based on programmable thresholds. All data is streamed in real time to the Blynk IoT cloud platform, providing a responsive web and mobile dashboard for remote monitoring and manual override.

The system operates in two modes: Auto Mode, where actuators are driven entirely by sensor readings and user-defined thresholds, and Manual Mode, where each actuator is controlled directly from the Blynk dashboard. An LED serves as a local visual indicator whenever any actuator is active.

Project Objectives

• Monitor temperature, humidity, and soil moisture in real time.
• Automatically activate the exhaust fan when temperature exceeds a set threshold.
• Automatically trigger the water pump when soil moisture drops below a set level.
• Provide a live cloud dashboard with historical graphs and manual controls.
• Allow threshold adjustment remotely without reflashing firmware.

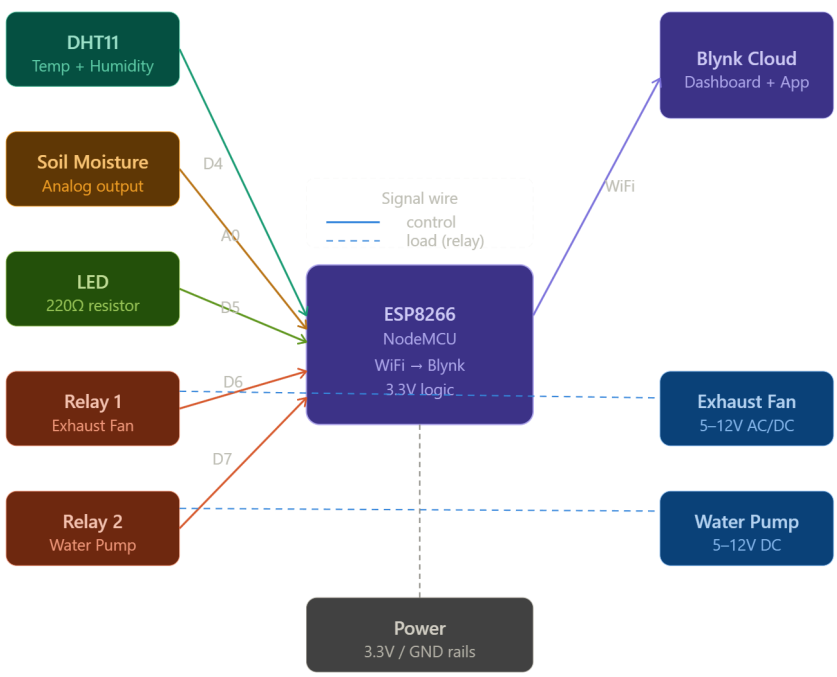
Components & Hardware

Component	Qty	Role
ESP8266 Node MCU	1	Main controller + Wi-Fi
DHT11 Sensor	1	Temperature & humidity
Soil Moisture Sensor	1	Soil dryness detection
5mm LED	1	Activity indicator
220 Ohm Resistor	1	LED current limiting
10kOhm Resistor	1	DHT11 pull-up
Relay Module	2	Fan and pump switching
Exhaust Fan	1	Ventilation / cooling
Water Pump	1	Irrigation
Jumper Wires	-	Connections
Breadboard	1	Prototyping
5V Power Supply		Node MCU Power

System Architecture & Logic

The firmware runs on a non-blocking event loop using the BlynkTimer library. Sensor reads and cloud updates are executed every 2 seconds without blocking the Blynk connection handler. The control logic is separated from the communication layer for reliability.

Condition	Action	Blynk Update
Temperature ≥ Threshold (°C)	Relay 1 → LOW (Fan ON)	Relay 1 → LOW (Fan ON)
Temperature < Threshold	Relay 1 → HIGH (Fan OFF)	V3 = 0
Soil Moisture % < Threshold	Relay 2 → LOW (Pump ON)	V4 = 1, V9 = 'Pump ON'
Soil Moisture % ≥ Threshold	Relay 2 → HIGH (Pump OFF)	V4 = 0
Fan OR Pump active	LED → HIGH (ON)	V5 = 1
Both OFF	LED → LOW (OFF)	V5 = 0, V9 = 'All OK'
Manual Mode enabled	Relays follow V3/V4/V5	All manual buttons active



Firmware

Required Libraries (Arduino IDE)

- Blynk
- DHT sensor library
- Adafruit Unified Sensor
- ESP8266 Board Package

Key Firmware Sections

The firmware is structured into four main sections:

- Configuration block: Defines BLYNK_TEMPLATE_ID, auth token, Wi-Fi credentials, pin assignments, and default threshold values.
- sendSensorData(): Called every 2 s by BlynkTimer. Reads DHT11 and soil ADC, maps raw values to percentages, applies auto-control logic, and pushes all values to Blynk virtual pins.
- BLYNK_WRITE handlers: Six handler functions (V3–V8) respond to dashboard input. Threshold sliders update variables live; mode toggle switches between auto and manual.
- setup() / loop(): Initialises GPIO pins to safe states, starts DHT11 and Blynk, then runs Blynk.run() + timer.run() continuously.

Future Applications

The smart greenhouse system developed in this project forms a solid foundation that can be extended far beyond a hobbyist setup. As IoT technology continues to mature and hardware costs decline, the principles demonstrated here are directly applicable to a wide range of real-world scenarios.

- **Large-Scale Commercial Farming** The core architecture of this system — sensor nodes communicating with a cloud dashboard — scales naturally to commercial greenhouse operations. A network of ESP8266 or ESP32 nodes can be deployed across multiple growing zones, each reporting independently to a central Blynk or custom MQTT broker. Farm managers can monitor hundreds of sensors from a single dashboard, set zone-specific thresholds for different crops, and receive instant alerts when conditions deviate. Integrating CO2 sensors, EC (electrical conductivity) meters for nutrient monitoring, and pH probes would make the system suitable for professional hydroponic and aeroponic farms.
- **Vertical Farming** Urban vertical farms, where crops are grown in stacked layers under artificial lighting, demand precise environmental control at every tier. The relay-based control logic in this project can be extended to manage LED grow lights on per-layer schedules, multi-zone ventilation, and drip irrigation circuits for each rack. With vertical farming gaining traction in cities across India and globally, affordable IoT-based control systems like this one offer a practical path to energy-efficient, high-yield urban food production.
- **Precision Agriculture in Open Fields** With minor modifications — replacing the wired soil sensor with a wireless LoRa-based node and adding a solar panel for power — the system can be deployed in open agricultural fields. Farmers can monitor soil moisture across large plots and trigger irrigation pumps remotely, reducing water waste significantly. Organisations like ICAR (Indian Council of Agricultural Research) and various agricultural universities are actively researching such deployments for smallholder farms.
- **Hydroponics and Aquaponics** In hydroponic systems, where plants grow in nutrient-enriched water rather than soil, the water pump control logic can be repurposed to manage nutrient dosing pumps and recirculation cycles. Combining this with a pH and EC sensor creates a closed-loop nutrient management system. Aquaponics setups — which combine fish farming with plant cultivation — additionally benefit from water temperature monitoring and automated aeration control, both of which are straightforward extensions of this project.
- **Smart Home Plant Care** On a smaller scale, the same system can be miniaturised into a smart indoor plant care device. A single NodeMCU, a soil sensor, and a small peristaltic pump can

automatically water potted plants and notify the homeowner via the Blynk app when soil is dry or water reservoir levels are low. This has strong commercial potential as a consumer IoT product.

- **Educational and Research Applications** This project is well suited as a teaching platform in engineering colleges and polytechnic institutes. Students gain hands-on exposure to embedded systems, sensor interfacing, relay-driven actuator control, cloud connectivity, and real-time data visualisation — all within a single project. Research institutions can extend the platform to study plant growth responses to precisely controlled temperature, humidity, and irrigation cycles, generating data sets for agronomic studies.
- **Disaster-Resilient Food Security Systems** In regions prone to drought or erratic rainfall, automated greenhouse systems can act as controlled micro-environments for food production. Solar-powered versions of this system, with local SD card data logging as a fallback when internet connectivity is unavailable, can operate reliably in remote or under-served areas, contributing to food security initiatives.

Conclusion

This project successfully demonstrates the design and implementation of an IoT-based smart greenhouse monitoring and control system using widely available, low-cost hardware. The ESP8266 NodeMCU serves as an effective platform for Wi-Fi-connected embedded applications, and the Blynk IoT platform provides a capable cloud backend and dashboard without requiring custom server infrastructure.

The system achieves its core objectives: continuous environmental sensing, threshold-based automatic actuation, real-time cloud telemetry, and remote manual override — all within a robust, non-blocking firmware architecture. The dual Auto/Manual operating mode makes the system practical both as a fully autonomous controller and as a remotely supervised tool.

From a broader perspective, the project illustrates how accessible IoT development has become. Components costing under ₹1,000 in total, combined with free cloud services and open-source libraries, are sufficient to build a system that would have required expensive proprietary hardware and software a decade ago. This democratisation of embedded development opens significant opportunities for students, farmers, researchers, and entrepreneurs alike.

The limitations of the current implementation — single-point sensor coverage, dependence on continuous Wi-Fi connectivity, and a resistive soil sensor that degrades over time — are well understood and provide clear direction for future work. Addressing these through multi-node mesh networks, offline data buffering, and capacitive soil sensors would bring the system to a level suitable for semi-professional deployment.

References

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